Engineering Mathematics - I · Unit_03 · Unit 3

DRAFT — UE26MA141A · Unit_03 · Unit 3 · Lagranges Linear equation continued

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Lagranges Linear equation continued

1 Lagranges Linear equation continued

Study focus: Lagranges Linear equation_continued. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000058 · EV0000059

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In the last in the last class, we solved problems on Lagrangius linear equation using method of grouping.

Key details

- So in this session, we'll solve problems on Lagrangius linear equation by using the method of multipliers as well as.
- As we can see this is capital P.
- And this is small p and this is q and this is small q this is r after choosing p q r the axillary equations are dx over capital p, dy over capital q and dz by r.
- It is easy to see that we can't apply the method of grouping for this case because we are not able to separate the variables.
- In order to proceed with the method of multipliers, we need to choose element.
- They are either constants or functions of xyz.
- So to find the first solution, we choose a multipliers as x, y, and z.
- Then each fraction of this expression 1 becomes x into dx plus y into dy plus z into dz.
- It means we multiply this dx by x, dy by y and dz by z.
- Even in the denominator, we multiply this capital P by X.
- capital Q by Y and R by Z.
- And it is easily integrable.
- Therefore, the first solution for this case is x square plus y square plus z square equals c1.
- In a similar manner, we choose another set of multipliers, namely LMN.
- In this case, again we get the denominator as 0.
- LMN are concept.
- Lx plus MY plus Lz equals C2.

- This is nothing but V of XYZ equals C2.
- So this is how we can apply method of multipliers for in order to solve Lagrangius linear equation.
- X of Z square minus Y square.
- This is small p.
- Y of x square minus z square, which is capital Q, this is small q, and this is r.
- Again, it is easy to see that we can't apply method of grouping, so we go for method of multipliers.
- By choosing multipliers as xyz, then each fraction of one becomes x into dx plus y into dy plus z into dz.

Example or application from the lecture

- Let us solve the problem Mz minus N y times douz by doux plus nx minus Lz times douz by dou y equals L y minus mx.
- and V of XYZ equals C2 and hence the general solution for the given problem is p of u comma v equals 0.
- Let's look at another problem.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000058 · EV0000059 · EV0000060 · +1 source units in pack.json

Core explanation · 05:00–09:59

Obviously in the denominator all six term gets cancel out.

Key details

- U of x y z equals c one to be x square plus y square plus z square equals c one.
- and plus x square and minus x square.
- This is another set of multipliers for finding the second which is nothing but v of x y z equals c2.
- For convenient purpose, instead of C2 we write log because all terms on the left hand side containing log.
- then xy z becomes c2.
- It is very easy to see that to find the first solution we can go for second and third ratio.
- So on integration we get y by z equals c1 because we get log y.
- equals log z plus log c one.
- Log a minus log b is log a by b so based on that we get y by z equals c one.
- So we are going for method of multipliers.
- So in this problem we apply both method of grouping as well as method of multipliers.
- Obviously by two by using the second and third fractions or ratios.

Example or application from the lecture

- It is easier to see from the given problem if x square if x is not there, you can cancel z square, or if y is not there and z is not there, you can cancel z square and minus z square and minus y square plus y square.

- Let's take up another problem y square plus z square of p minus x y q plus zx equals 0.
- And hence the general solution is given by phi of y by z x square plus y square plus z square equals zero.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000062 · EV0000063 · EV0000064 · +1 source units in pack.json

Core explanation · 10:00–14:59

We get dy by y and dz by z so on integration.

Key details

- Should we go for grouping again or we can go for multipliers or we should go for differential concept.
- Even multipliers doesn't work.
- As from theory, this is similar to method of multipliers.
- LP plus MQ plus Nr.
- This is equated with a fraction of this is dx by p or dy by q or dz by r in order to get the integral.
- It means once we achieve this, we're going to equate this either with the first one or second one or third one.
- Depends on the problem.
- That is differential concept.
- So it means how do we do simplification?
- plus z divided by so multiply this by x so that gives x power three minus xy square minus x z square multiply second by y so we'll get a two x y square.
- The last one by z we get 2x square.
- Again same numerator.
- In the denominator you can take x common.
- It could be first one, second one, or third one.
- So unfortunately for this problem we cannot equate this with the first fraction because it doesn't have x.
- So we can equate this either with the second one or third one because it has got a single x here.
- We should observe carefully.
- is the exact differential of denominator.
- In the denominator, as usual, we have x square plus y square plus z square.
- On the right hand side we have dz by z.
- Observe the left hand side carefully, it is like the numerator is the exact differential of denominator.
- So the integration of this gives log of x square plus y square plus z square and right hand side we get log z plus log b for convenient purpose because other two terms are log.
- Therefore, if you use log a plus log b we get the solution as x square plus y square plus z square divided by z equals b.
- So in this problem to get the first solution we use the as usual grouping method.

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EV0000066 · EV0000067 · EV0000068 · +1 source units in pack.json

Core explanation · 15:00–19:59

To get the second solution we use the differential concept method.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000070 · EV0000071 · EV0000072 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Lagranges Linear equation_continued transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/004_Lagranges_Linear_equation_continued__7d0c7f14ab47b67a.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Lagranges Linear equation_continued transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/004_Lagranges_Linear_equation_continued__7d0c7f14ab47b67a.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Lagranges Linear equation_continued transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/004_Lagranges_Linear_equation_continued__7d0c7f14ab47b67a.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9992; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Lagranges Linear equation_continued transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/004_Lagranges_Linear_equation_continued__7d0c7f14ab47b67a.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
2. 1 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
3. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
4. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
5. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
6. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
7. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00022	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0054. Sidecar variants compare...
SRC00023	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0054. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00024	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0054. Sidecar variants compare...
SRC00025	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0054. Sidecar variants compare...

C Revision and retrieval practice